Inferences about Population Variances

Confidence Interval for Population VariancesEstimation and Tests for Comparing Two Population Variances

Inferences about population variances

Estimation and Tests for a Single Population Variance

- Recall that the sample variance $s^2 = \sum(y - \bar{y})^2/(n-1)$ is the *point estimate* of σ^2.
- For tests and confidence intervals about σ^2 we use the fact that the sampling random variable
  $(n-1)\,S^2/\sigma^2 = \chi^2$
  has the *Chi-square Distribution* with $n-1$ degrees of freedom or df, when the sample is from a normal population.

- The chi-square distribution is nonsymmetric.
- Like the Student's t distribution, there is a different curve for each sample size n i.e., for each value of *df*.
- The distribution appear to be more skewed for smaller values of df and become more symmetric as df increases.
- Because the chi-square distribution is nonsymmetric, the percentiles for probabilities at both ends needs to be tabulated.

Chi-square Distribution

A $100(1-\alpha)\%$ Confidence Interval for σ^2

This has the form:

$$\frac{(n-1)s^2}{\chi^2_U} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_L}$$

where

- Since $df = n-1$ look up χ^2_{n-1} percentiles.
- χ^2_L is the lower-tail value with area $\alpha/2$ to the left.
- χ^2_U is the upper-tail value with area $\alpha/2$ to the right.
- The confidence interval for the standard deviation σ is found by taking square roots of both end points of above.

Example

Let's assume that the normal probability plot of the data appear to show that the sample is from a normal distribution. From the data $n = 30$, $\bar{y} = 500.453$, $s = 3.433$ were calculated.

A 99% confidence interval for σ^2 is computed as follows:

Since $\alpha = .01$, $\alpha/2 = .005$ and $1 - \alpha/2 = .995$, we compute

$\chi^2_L = \chi^2_{0.995,29} = 13.12 \quad \chi^2_U = \chi^2_{0.005,29} = 52.34$

Thus the endpoints for the required C.I. are, respectively,:

$\frac{(n-1)s^2}{\chi^2_U} = \frac{29(3.433^2)}{52.34} = 6.53 \; \frac{(n-1)s^2}{\chi^2_L} = \frac{29(3.433^2)}{13.12} = 26.05$

Thus $(6.53, \; 26.05)$ is a 99% C.I. for σ^2

Example

Taking square roots of both endpoints of the interval for σ^2:
we get $(2.56,\ 5.10)$ as a 99% C.I. for σ.
Thus we are 99% confident that the standard deviation of the weights of coffee containers lies in the interval $(2.56,\ 5.10)$.
The filling machine was designed to be so that $\sigma = 4$ grams, and since this value falls in the above interval, the machine satisfies the design specification for standard deviation.
A separate confidence interval for μ is computed for checking whether the design specification for the mean is satisfied.

Example: R

$$\frac{(n-1)s^2}{\chi^2_U} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_L}$$

```r
n <- 30; ybar <- 500.543; s <- 3.433; alpha <- 0.01

# CI for sigma^2
sigma2CI <- (n-1)*s^2 / qchisq(c(1-alpha/2, alpha/2), df=n-1)
sigma2CI
```

```
[1]  6.530527 26.047961
```

```r
# CI for sigma
sqrt(sigma2CI)
```

```
[1] 2.55549 5.10372
```

Tests of Hypotheses about σ^2

Test:

1. $H_0 : \sigma^2 \leq \sigma_0^2$ vs. $H_a : \sigma^2 > \sigma_0^2$
2. $H_0 : \sigma^2 \geq \sigma_0^2$ vs. $\sigma^2 < \sigma_0^2$
3. $H_0 : \sigma^2 = \sigma_0^2$ vs. $\sigma^2 \neq \sigma_0^2$

Test Statistic: $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$

Rejection Region: for specified α and $df = n - 1$,

1. Reject H_0 if $\chi^2 > \chi^2_U$, where $\chi^2_U \equiv \chi^2_{\alpha,\, n-1}$
2. Reject H_0 if $\chi^2 < \chi^2_L$, where $\chi^2_L \equiv \chi^2_{1-\alpha,\, n-1}$
3. Reject H_0 if $\chi^2 > \chi^2_U$, where $\chi^2_U \equiv \chi^2_{\alpha/2,\, n-1}$, or if $\chi^2 < \chi^2_L$, where $\chi^2_L \equiv \chi^2_{1-\alpha/2,\, n-1}$

Notes about Example

- The above inferences are based on the assumption of sampling from a normal population.
- They are more sensitive to departures from normality than the inferences about population mean.
- We don't have a CLT-type theorem which applies to S^2 like we have in case of $\bar{Y}$.
- Always plotting sample data as a preliminary procedure - using boxplot or normal probability plot to look for skewness, or outliers, is recommended.

Estimation and Tests for Comparing Two Population Variances

- We will consider the case of independent samples from two populations having variances σ_1^2 and σ_2^2.
- The main application is testing $\sigma_1^2 = \sigma_2^2$.
- Theory says that when the two populations are Normal, and sample sizes are n_1 and n_2 then the random variable

  $$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$$

  has the F distribution with degrees of freedom $df_1 = n_1 - 1$ and $df_2 = n_2 - 1$.

Estimation and Tests for Comparing Two Population Variances - cont.

- df_1 is the degrees of freedom associated with S_1^2.
- df_2 is the degrees of freedom associated with S_2^2.
- The F distribution is a two parameter family indexed by df_1 and df_2.
- Some use the terms Numerator and Denominator degrees of freedom for df_1 and df_2.
- The F-statistic is calculated as

$$F_c = \frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2}$$

Estimation and Tests for Comparing Two Population Variances - cont.

- The lower tail values, when needed can be obtained by the relationship
  $F_{1-\alpha,df_1,df_2} = \dfrac{1}{F_{\alpha,df_1^*,df_2^*}}$
  where $df_1^* = df_2$ and $df_2^* = df_1$.
- That is, to calculate a left tail probability, look-up the corresponding right tail probability after switching the numerator and denominator degrees of freedom.
- For example $F_{.95,3,10}$ is calculated using the above relationship as
  $1/F_{.05,10,3} = 1/8.79 = 0.11$
- And $F_{.95,10,3}$ is calculated using the above relationship as
  $1/F_{.05,3,10} = 1/3.71 = 0.27$

Tests comparing two variances

Test:

1. $H_0 : \sigma_1^2 \leq \sigma_2^2$ vs. $H_a : \sigma_1^2 > \sigma_2^2$
2. $H_0 : \sigma_1^2 = \sigma_2^2$ vs. $H_a : \sigma_1^2 \neq \sigma_2^2$

Test Statistic:

$$F = \frac{s_1^2}{s_2^2}$$

Rejection Region: For given α, $df_1 = n_1 - 1$, and $df_2 = n_2 - 1$.

1. Reject H_0 if $F \geq F_{\alpha, df_1, df_2}$
2. Reject H_0 if $F \leq F_{1-\alpha/2, df_1, df_2}$ or $F \geq F_{\alpha/2, df_1, df_2}$

Example

In order to test the hypothesis $H_0 : \mu_1 - \mu_2 = 0$ using a two-sample t-statistic with pooled sample variance, we need to assume $\sigma_1^2 = \sigma_2^2$.
To check whether this is a valid assumption, let us formally test
$H_0 : \sigma_1^2 = \sigma_2^2 \; vs. \; H_a : \sigma_1^2 \neq \sigma_2^2$
using $\alpha = .05$
The two independent samples gave sample variances:
$s_1^2 = 0.105, \quad s_2^2 = 0.058$ with $df_1 = 9$ and $df_2 = 9$.

Example - cont.

Test: $H_0 : \sigma_1^2 = \sigma_2^2 \; vs. \; H_a : \sigma_1^2 \neq \sigma_2^2$

Test Statistic: $F_c = \frac{s_1^2}{s_2^2} = \frac{0.105}{0.058} = 1.81$

Rejection Region: $F \leq F_{.975,9,9}$ or $F \geq F_{.025,9,9}$

From Table 8, using $\alpha = 0.05$, $df_1 = 9, df_2 = 9$, we have

$F_{0.025,9,9} = 4.03$ and $F_{0.975,9,9} = 1/F_{0.025,9,9} = 1/4.03 = 0.25$

Thus we reject if $F \leq 0.25$ or $F \geq 4.03$.

Since $F_c = 1.81$ does not fall in the rejection region, we fail to reject $H_0 : \sigma_1^2 = \sigma_2^2$.

Conclusion: The assumption of equal variances for the two populations appears reasonable.

Example: R

$$H_0 : \sigma_1^2 = \sigma_2^2 \; vs. \; H_a : \sigma_1^2 \neq \sigma_2^2$$

```r
s12 <- 0.105; s22 <- 0.058; df1 <- 9; df2 <- 9; alpha <- 0.05
f <- s12/s22
f
```

```
[1] 1.810345
```

```r
# Rejection region
rr <- qf(c(alpha/2, 1-alpha/2), df1=df1, df2=df2)
rr
```

```
[1] 0.2483859 4.0259942
```

```r
# f in rejection region?
f<rr[1] | f>rr[2]
```

```
[1] FALSE
```

Example - cont.

What if we are instead interested in the test

$$H_0 : \sigma_1^2 \leq \sigma_2^2 \; vs. \; H_a : \sigma_1^2 > \sigma_2^2$$

```r
s12 <- 0.105; s22 <- 0.058; df1 <- 9; df2 <- 9; alpha <- 0.05
f <- s12/s22
f
```

```
[1] 1.810345
```

```r
# Rejection region
rr <- qf(1-alpha, df1=df1, df2=df2)
rr
```

```
[1] 3.178893
```

```r
# f in rejection region?
f > rr
```

```
[1] FALSE
```

```r
# p-value
pf(f, df1=df1, df2=df2, lower.tail = F)
```

```
[1] 0.1949263
```

Using var.test()

Say you have a dataset containing data with samples from two populations. We can use the `var.test()` function to perform the previous test.

$H_0 : \sigma_1^2 \leq \sigma_2^2 \; vs. \; H_a : \sigma_1^2 > \sigma_2^2$

```r
set.seed(412)
x <- rnorm(n=20, mean = 10, sd = 8)
y <- rnorm(n=20, mean = 20, sd = 5)

boxplot(c(x,y)~rep(c('x', 'y'), each = 20))
```

```r
qqnorm(x); qqline(x)
```

```r
qqnorm(y); qqline(y)
```

```r
s12 <- var(x); s22 <- var(y); df1 <- length(x)-1; df2 <- length(y)-1; alpha <- 0.05
f <- s12/s22
f
```

```
[1] 2.308348
```

```r
pf(f, df1=df1, df2=df2, lower.tail = F)
```

```
[1] 0.03799438
```

```r
var.test(x, y, alternative = 'greater')
```

```
    F test to compare two variances

data:  x and y
F = 2.3083, num df = 19, denom df = 19, p-value = 0.03799
alternative hypothesis: true ratio of variances is greater than 1
95 percent confidence interval:
 1.064613      Inf
sample estimates:
ratio of variances
          2.308348
```

Test comparing several variances

Hartley's F_{max} Test for Homogeneity of Variances

This test requires that independent random samples of equal samples sizes, n, are drawn from t populations having normal distributions.

Test: $H_0 : \sigma_1^2 = \sigma_2^2 = \ldots = \sigma_t^2$ vs. H_a : Not all σ^2's equal

Test Statistic: $F_{max} = \frac{s_{max}^2}{s_{min}^2}$ where s_{max}^2 =largest sample variance and s_{min}^2 =smallest sample variance

Rejection Region: For given α, reject H_0 if F_{max} exceeds the tabulated percentile

Test comparing several variances cont.

- When sample sizes are not all equal an approximate test is obtained if n is replaced with n_{max} , where n_{max} is largest sample size, in the above procedure.
- Hartley's test is quite sensitive to violation of the normality assumption.
- Thus other procedures that do not require the normality assumption, such as Levene's test has been proposed.
- Levene's test is too cumbersome to calculate by hand; so software may have to be used.
- Levine's test procedure is less powerful than Hartley's test when populations have normal distributions.

Speaker notes